

Pin-Jun Chen

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RESEARCH INTERESTS

3D Integrated Circuits: enhance the performance of TSV-based/monolithic 2.5D/3D ICs by electrical and thermal design.

AI Hardware: leverage heterogeneous integration to realize scalable, energy-efficient, and high-speed chips for AI/ML.

EDUCATION BACKGROUND

Georgia Institute of Technology (Georgia Tech) (GPA: 3.83/4.00, 95.75%)

Atlanta, GA

PhD in Electrical and Computer Engineering

August 2024 – Expected August 2029

- **Research Advisor:** Dr. Shimeng Yu, Full Professor, Georgia Tech.
- **Relevant Coursework:** Microelectronics Systems Packaging (A), Hardware Software Co-Design for Machine Learning Systems (A), Machine Learning (A), Advanced Programming Techniques (A), Memory Device Technologies (A)

Katholieke Universiteit Leuven (KU Leuven)

Leuven, Belgium

International Scholar, Faculty of Science

January 2020 – May 2020

- Granted admission as an International Scholar in the Faculty of Science at KU Leuven, with comprehensive sponsorship covering a living stipend, health insurance premiums, tuition support, and travel expenses.

National Chiao Tung University (NCTU) (GPA: 4.07/4.30, 91.57%)

Hsinchu, Taiwan

MSc in Semiconductor Technology

September 2017 – May 2020

- **Research Advisor:** Dr. Kuan-Neng Chen, Chair Professor, NCTU
- **Master's Thesis:** "Process Development and Simulation Analysis for Lowering Thermal Budget on Stacked/Monolithic 3D Integrated Circuits"
- **Relevant Coursework:** Semiconductor Processing (A+), Semiconductor Physics and Devices (A), Nanometer Scale Electronic Devices (A), Reliability and Failure Physics of Semiconductor Devices (A-), 3D Integrated Circuit (A-)

The Chinese University of Hong Kong (CUHK)

Hong Kong, China

Visiting Student, Department of Chemistry

July 2017 – August 2017

- Selected as a summer research assistant at Prof. Chit Tsui's lab in the Department of Chemistry.

National Tsing Hua University (NTHU) (GPA: 3.58/4.30, 82.50%)

Hsinchu, Taiwan

BSc in Materials Science and Engineering

September 2013 – June 2017

- **Relevant Coursework:** Electromagnetism (A+), Engineering Mathematics (A+), Electronics Laboratory (A+), Calculus (A), The Physical Properties of Materials (A), Introduction to Quantum Physics (A-)

PUBLICATIONS & CONFERENCE PRESENTATIONS ([Google Scholar](#))

- Wei-Hsing Huang, Janak Sharda, Cheng-Jhih Shih, Yuyao Kong, Faaqi Waqar, **Pin-Jun Chen**, Yingyan (Celine) Lin, and Shimeng Yu, "A3D-MoE: Acceleration of Large Language Models with Mixture of Experts via 3D Heterogeneous Integration," *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, July 2026, doi: 10.1109/JXCDC.2026.3713957
- **Pin-Jun Chen**, Janak Sharda, Po-Kai Hsu, Min Gyu Park, Sung Kyu Lim, Muhannad Bakir, and Shimeng Yu, "3D Stacked GPU Architecture with Ultra-Thin 3D DRAM Dies and Voltage Regulators Embedded into Glass Substrate," in *IEEE 76th Electronic Components and Technology Conference (ECTC)*, Orlando, FL, May 26-29, 2026, doi: 10.1109/ECTC51846.2026.00241. (oral presentation)
- Tzu-Chieh Chou, Shin-Yi Huang, **Pin-Jun Chen**, Han-Wen Hu, Demin Liu, Chih-Wei Chang, Tzu-Hsuan Ni, Chao-Jung Chen, Yu-Min Lin, Tao-Chih Chang, and Kuan-Neng Chen, "Electrical and reliability investigation of Cu-to-Cu bonding with silver passivation layer in 3D integration," *IEEE Transactions on Components, Packaging and Manufacturing Technology (TCPMT)*, vol. 11, no. 1, pp. 36-42, Nov. 2020, doi: 10.1109/TCPMT.2020.3037365.
- Ping-Yi Hsieh, Yi-Jui Chang, **Pin-Jun Chen**, Chun-Liang Chen, Chih-Chao Yang, Po-Tsang Huang, Yi-Jing Chen, Chih-Ming Shen, Yu-Wei Liu, Chien-Chi Huang, Ming-Chi Tai, Wei-Chung Lo, Chang-Hong Shen, Jia-Min Shieh, Da-Chiang Chang, Kuan-Neng Chen, Wen-Kuan Yeh, and Chenming Hu, "Monolithic 3D BEOL FinFET switch arrays using the location-controlled-grain technique in voltage regulator with better FOM than 2D regulators," in *IEEE 65th International Electron Devices Meeting (IEDM)*, San Francisco, CA, Dec. 7-11, 2019, pp. 3.1.1-3.1.4, doi: 10.1109/IEDM19573.2019.8993441.

- **Pin-Jun Chen**, Chih-Ming Shen, Chih-Chao Yang, Ming-Chi Tai, Wei-Chung Lo, Chang-Hong Shen, Chenming Hu, and Kuan-Neng Chen, "Transient thermal damage simulation for novel location-controlled grain technique in monolithic 3D IC," in *IEEE 14th International Microsystems, Packaging, Assembly and Circuits Technology Conference (IMPACT)*, Taipei, Taiwan, Oct. 23-25, 2019, pp. 104-107, doi: 10.1109/IMPACT47228.2019.9024960. **(oral presentation)**

HONOURS & AWARDS

- **2026 Taiwan Government Scholarship for Study Abroad** (a yearly scholarship of USD 30,000 for a max duration of three years provided by Taiwan Ministry of Education)
- **2025 Indigenous People Scholarship for Overseas Study** (a monthly subsidy of USD 500 for a max duration of three years provided by Taiwan Council of Indigenous Peoples)
- **2020 KU Leuven Scholarship** (one-year full scholarship of USD 13,500 sponsored by KU Leuven)
- **2019 Future Tech Breakthrough Award** (our joint research work won the winners list of the Future Tech Breakthrough Award held by the Taiwan Ministry of Science and Technology)
- **2019 Taiwan Semiconductor Manufacturing Company (TSMC) Fellowship** (one-year fellowship of USD 2,300 sponsored by TSMC for outstanding research performance)
- **2019 Helm Technology Scholarship** (one recipient for each college in NCTU for exceptional academic work)
- **2018 Helm Technology Scholarship**
- **2017 The University System of Taiwan-CUHK Scholarship**

RESEARCH & WORK EXPERIENCE

Georgia Institute of Technology

Atlanta, GA

Graduate Research Assistant, School of Electrical and Computer Engineering

August 2024 – Present

- Design advanced 3D architectures to improve power, performance, and area (PPA) parameters for AI accelerators.
- Conducted electro-thermal co-analysis and device simulation to characterize emerging memory technologies for AI/HPC applications.

Ericsson

Taipei, Taiwan

5G Early Career Program Trainee, Market Area Northeast Asia

November 2020 – March 2023

- Selected as one of the seven elite trainees from a diverse pool of candidates across Taiwan, China, Hong Kong, Macau, Japan, and South Korea, to advance regional 5G initiatives.
- Promoted Ericsson's cloud radio access network in IEEE GLOBECOM 2020, one of the IEEE Communications Society's flagship conferences dedicated to driving innovation in nearly every aspect of communications.
- Led and managed the projects, provided solutions for Asia Pacific Telecom and Far EastOne Telecom in 4G/5G network deployment (LTE 700M/LTE 2.6G, NR 28G), establishing Taiwan's first 5G non-standalone multi-operator core network for increased throughput and improved subscriber experiences, particularly in rural and mountainous areas.

Interuniversity Microelectronics Centre (imec)

Leuven, Belgium

Research and Development Intern, Wafer Bonding and Assembly

January 2020 – May 2020

- Operated a laminator, bonder, and SAM to explore the optimal parameters for the dry film photoresist (DFR) process.
- Fabricated 3D microfluidic devices using the DFR technique with novel processes implemented.

Qualcomm

Hsinchu, Taiwan

Hardware Engineering Intern, Reliability Engineering

May 2019 – November 2019

- Acquired knowledge in product-level reliability evaluation for cutting-edge silicon processes used in mobile products.
- Learned to use statistical analysis techniques such as Weibull and Lognormal distributions, reliability tools or software, high-temperature-operating-life ovens, and electrostatic discharge machines.

NCTU-TSMC Joint Research Center

Hsinchu, Taiwan

Research Assistant, Center for Advanced Semiconductor Technology Research

September 2017 – September 2019

- Developed the next-generation monolithic back-end-of-line 3D FinFETs in collaboration with Taiwan Semiconductor Research Institute and NCTU-Berkeley Joint Research Center.
- Designed the laboratory's first thermal simulation environment for monolithic 3D ICs.
- Utilized ANSYS to analyze transient thermal conduction in various FinFET models under different boundary conditions.

Industrial Technology Research Institute*Research Intern, Electronics and Optoelectronics Research Laboratories***Hsinchu, Taiwan***May 2018 – April 2019*

- Employed various materials as passivation layers to improve the Cu-to-Cu bonding technique.
- Enhanced the manufacturability of TSV-based 3D stacked ICs by reducing the thermal budget in processes.

TECHNICAL SKILLS & LANGUAGES

- **Integrated Circuits Fabrication:** Photolithography, Sputtering, Wafer Bonding, Dry Film Lamination, Spin Coating, Liftoff
- **Material Analysis:** Scanning Electron Microscope, Energy Dispersive Spectroscopy, Scanning Acoustic Microscope
- **Computer Skills:** ANSYS Workbench/HFSS, Sentaurus TCAD, Cadence Innovus/Virtuoso, C/C++, Python, MATLAB
- **Languages:** Mandarin (Native), Taiwanese (Native), Hakka (Native), and English (Proficient)